

VALIDATION REPORT OF HYDROELECTRIC PROJECT EL EDÉN



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Summary:

ICONTEC was contracted for the validation of Hydroelectric Project El Edén, this validation process was performed under VCS standard version 3.3. The Hydroelectric Project El Edén is located in the departament of Caldas, Colombia, this project activity will be developed in the municipalities of Pensilvania, Manzanares and Marquetalia. The opening meeting between ICONTEC and El Eden Central Hidroelectrica S.A.S. E.P.S. was carried out after the project was listed in the VCS project pipeline on May 22nd, 2013.

“Hydroelectric Project El Edén” consists of the construction of a new hydroelectric power plant that takes advantage of the La Miel River. The project will have and installed capacity of 20.23 MW and a projected annual average generation of 104.5 Gigawatt hours (GWh). The purpose of the project is the electricity generation using renewable energy sources to supply the Colombian National Interconnected Electric System (SIN).

This project proposes to apply baseline and monitoring methodology identified as ACM0002, version 13.0.0 of Approved methodologies for large scale CDM project activities by the CDM Executive Board. This methodology has been accepted to be used by the VCS standard.

The purpose of this validation process is to confirm the compliance of the project with the VCS standard (version 3.3), based on the PD version 04 of 16/09/2013, and the proper application of the monitoring methodology ACM0002 (Version 13.0.0) and its related CDM tools and guidelines.

During this validation process, 11 findings occurred to be classified as 6 corrective actions request and 5 clarifications actions request, which were treated by the project owner and clarified in a new version of the PD version 04. Upon review of the documentation and explanations provided by the Project Proponent, all findings were closed out in a clear and transparent manner.

ICONTEC verified the project design and the implementation status through the documental review and onsite visit (it was conducted on May 28th, 2013 to May 30th, 2013), addressing conservatively the restrictions and uncertainties associated to this validation process.

The validation team was able to conclude that as it was described in the version 04 of the project description, it meets all relevant VCS requirements and correctly applies the baseline and monitoring methodology ACM0002, version 13.0.0. Hence, ICONTEC requests the registration of the project.

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1 INTRODUCTION

1.1 Objective

The purpose of a validation is to secure the opinion of an independent third party in order to assess the project's design: the project's baseline, the monitoring plan, and the project's compliance with relevant VCS requirements.

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of certified emission reductions (CERs).

1.2 Scope and Criteria

The validation scope involves an independent and objective review to determine that the project design meets the following VCS criteria:

- VCS Program Guide, version 3.4.
- VCS Standard, version 3.3.
- Program Definitions, version 3.4
- Program Fee Scheduled, version 3.3
- Approved Consolidated Baseline and Monitoring Methodology ACM0002, Version 13.0.0

ICONTEC carries out audits according to its ethics code and internal procedures for carrying out validation, verification and certification audits of VCS project activities, which, in turn, are based on the VCS Standard. Likewise, ICONTEC focuses on the identification of significant risks for CER generation, and verification of the mitigation during its audits.

The validation does not intend to provide any consulting for the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

1.3 Level of assurance

The information sources used are deemed reliable, these are: PD, on-site visit to the project proponent, on-site interviews, document review and secondary sources for confirmation.

Besides, the validation report is submitted to an internal review that is performed by a third party, who impartially reviews the report and issues an opinion regarding the report's reliability.

1.4 Summary Description of the Project

The project activity consists of the supply of clean hydroelectric energy to the Colombian National Interconnected System (SIN) through the implantation and operation of a Greenfield Hydro Power Plant (HPP). The Plant will have an installed turbine capacity of 20.23 MW and is expected to generate 104.5 GWh/year of clean and renewable electric energy. The Plant is expected to start its commercial operations on 01/01/2015.

In the absence of the project activity, the clean electricity supplied by Hydroelectric Project El Edén to the SIN would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

2 VALIDATION PROCESS

2.1 Method and Criteria

The validation consists of the following three phases:

- i) A desk review of the project design documents,
- ii) Follow up interviews with project stakeholders,
- iii) Resolution of outstanding issues and the issuance of a final validation report and opinion.

As mentioned in clause 1.2 of this report ICONTEC, based on its ethics code and internal procedures, carries out validation, verification and certification audits of VCS project activities (which, in turn, are based on the VCS standard) focused on the identification of significant risks for VER generation and the verification of the contribution to climate change mitigation.

All documentation review during the validation process has been including in section 2.2 – Document Review.

Findings established during the validation can be seen as:

- A non-fulfillment of validation protocol criteria, or
- An identified risk to the fulfillment of the project objectives

The findings could take the form of a Corrective Action Request (CAR), Forward Action Request (FAR) or a Clarifications Request (CL).

Corrective action requests (CAR) are issued where:

- i) The project participants have made mistakes which directly influence the ability of the project activity to achieve real, measurable and additional emission reductions;
- ii) The VCS requirements have not been met; or
- iii) There is a risk that emission reductions cannot be monitored or calculated

A Forward Action Request is made to highlight issues related to project implementation that will require review during the next verification of the project activity.

A Clarification Request is required when information is insufficient or not clear enough to establish whether a requirement has been met.

ICONTEC resolve or “close out” CARs and CLs only if the project participants modify the project description, rectify the PD or provide additional explanations or evidence that satisfy the ICONTEC’s concerns.

This validation report explains the issues raised, the responses provided by the project proponent, the means of validation of such responses and references to any resulting changes in the PD or supporting annexes.

2.2 Document Review

PD submitted by El Eden Central Hidroelectrica S.A.S. E.P.S. and the additional background documents related to the project design and baseline were assessed during the validation.

Main documents reviewed are:

- /1/ Project Description version 1 (issued on April 3rd, 2013), version 2 (issued on June 21st, 2013), version 2.1 (issued on June 26th, 2013), version 3 (issued on July 10th, 2013), and version 04 (issued on September 16th, 2013)
- /2/ Spreadsheet with emission reduction calculations (files: EF Colombia 2009-2011.xls and EF Colombia 2010-2012.xls)
- /3/ Spreadsheet with the financial calculation for additionality assessment (files: Equity IRR Hydroelectric Project El Edén (4Apr13).xls, Equity IRR Hydroelectric Project El Edén (30May13).xls, Equity IRR Hydroelectric Project El Edén (11Jun13).xls) and Equity IRR Hydroelectric Project El Edén (28Aug13)
- /4/ Technical Description about hydroelectric use of la Miel River – Project El Edén, issued by Latinco S.A. and I-Consult dated on December 2010
- /5/ Hydrologic Studies carried out by I-Consult to El Edén hydroelectric project, dated on May 30th, 2013
- /6/ Technical offer proposed by Andritz Hydro to Central Hidroeléctrica El Edén S.A.S E.S.P, for electromechanical equipment of El Edén hydroelectric project, dated on March 12th 2013.
- /7/ Environmental Impact Study about hydroelectric use of la Miel River – Project El Edén,

Executive Summary issued by Latinco S.A. and I-Consult dated on November 2010

- /8/ Resolution N° 173 issued by the CORPOCALDAS¹, dated on May 4th/2011, which grants the Environmental license to El Eden Hydroelectric project, this license was assigned to Central Hidroeléctrica El Edén S.A.S E.S.P
- /9/ Modification to Environmental Impact Study about the construction and operation of El Edén Hydroelectric Project issued by Latinco S.A. and I-Consult dated on November 2011
- /10/ Environmental Accomplishment Report (ICA) N° 1 issued by Central Hidroeléctrica El Edén S.A.S E.S.P and dated on December 22nd/2011 presented to CORPOCALDAS
- /11/ Environmental Accomplishment Report (ICA) N° 2 issued by Central Hidroeléctrica El Edén S.A.S E.S.P and dated on June 28th/2012 presented to CORPOCALDAS
- /12/ Environmental Accomplishment Report (ICA) N° 3 issued by Central Hidroeléctrica El Edén S.A.S E.S.P and dated on December 20th/2012 presented to CORPOCALDAS
- /13/ Letter issued by UPME² to Latinco S.A. dated on January 25th/2011 acknowledging the registration of the El Eden Hydroelectric Project as a generation project
- /14/ Construction Contract for the infrastructure implementation and operation of El Eden Hydroelectric project, signed between Central Hidroeléctrica El Edén S.A.S E.S.P and UT-GC CHOC³ and dated on April 16th/2012.
- /15/ Letter issued by UPME² dated on January 25th/2011, acknowledging the registration of the El Edén Hydroelectric Project as a future generation project to be connected to Colombian Electrical grid
- /16/ Energy purchase proposal made by Synermin about the energy production of El Edén Hydroelectric Project, dated on May 10th/2012
- /17/ Presentation performed on May 16th/2012 about the El Edén Hydroelectric Project development, when the shareholders of Central Hidroeléctrica El Edén S.A.S E.S.P took the decision to go ahead with project implementation.
- /18/ Report about Tariffs and commercial transactions on the Colombian Wholesale Market⁴, issued by XM⁵, dated on September 2011

¹ Regional Autonomous Corporation of Caldas (in Spanish: Corporación Autónoma Regional de Caldas)

² Colombian Energy and Mining Planning Unit (in Spanish: Unidad de Planeación minero Energética)

³ Temporary Union – Builder Group of Hydroelectric Plants in Eastern Caldas (in Spanish: Unión Temporal – Grupo Constructor Centrales Hidroelectricas Del Oriente de Caldas)

- /19/ Renewable Power Generation Costs in 2012, An Overview, issued by IRENA⁶, dated on 2013
- /20/ Quotation made by Andritz Hydro for electromechanical equipment for El Edén Hydroelectric project, issued on March 12th, 2013
- /21/ Spreadsheet with the common practice analysis (file: Common Practice Analysis El Edén 11Jun2013.xls)
- /22/ Reference Expansion Plan for Generation and Transmission 2010-2024, issued by UPME, dated on November 2010.
- /23/ Decree 3019 issued by the Colombian Ministry of Finance and Public Credit, dated on December 26th, 1989
- /24/ O&M Annual Budget for El Edén Hydroelectric Plant, dated on July 2013
- /25/ "IPCC, 2011: IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation", issued by the Working Group III of the Intergovernmental Panel on Climate Change, dated in May 2011. File "Chapter 5 Hydropower IPCC.pdf"

Background documents related to methodologies employed in the design or other reference document:

- /26/ ACM0002, Consolidated baseline methodology for grid-connected electricity generation from renewable sources, version 13.0.0.
- /27/ Tool to calculate the emission factor for an electricity system, version 03.0.0
- /28/ Tool for the demonstration and assessment of additionality, version 07.0.0
- /29/ Guidelines on the assessment of investment analysis, version 05.0
- /30/ Guidelines for the reporting and validation of plant load factors, version 01.0
- /31/ Guidelines on common practice, version 02.0

⁴ This report is available in
http://www.xm.com.co/Informe%20Mensual%20Analisis%20del%20Mercado/03_Informe_Precios_y_Transacciones_TXR_09_2011.pdf

⁵ Colombian Electrical Market Administrator and Operation Administrator

⁶ This report is available on
http://www.irena.org/DocumentDownloads/Publications/Overview_Renewable%20Power%20Generation%20Costs%20in%202012.pdf

2.3 Interviews

The following table outlines the persons having been directly in contact with ICONTEC within the validation process:

Follow up interviews

Date	Interview Delegate	Organization and Role	Interview Topics
May 28 th /2013	Jose Bernardo Alvarez	Deputy Manager El Edén Central Hidroeléctrica S.A.S. E.P.S.	Participation, Project Description, Baseline and Monitoring Methodology, Monitoring Plan, Sustainable Development, Local Stakeholder Consultation, Environmental Impacts
May 29 th /2013	Carolina Mesa	Environmental Engineer I-Consult	Sustainable Development, Environmental Impacts
May 30 th /2013	Pablo Agudelo	Engineering Manager I-Consult	Baseline and Monitoring Methodology
May 29 th and 30 th /2013	Juan Pablo Angel	Manager El Eden Central Hidroeléctrica S.A.S. E.P.S.	Participation, Project Description, Baseline and Monitoring Methodology, Additionality, Monitoring Plan, Sustainable Development, Local Stakeholder Consultation, Environmental Impacts
	Christian Ehrat	Consultant	

2.4 Site Inspections

An onsite visit was carried out by the lead auditor on May 28th/2013. This visit included a tour by the future installations of the hydroelectric power plant (water intake, penstock, powerhouse and discharge channel). The project has only begun with civil works (no electrical equipment has been installed), hence at the moment of the visit it was not possible to confirm, with the sense of sight, the project's operation since the project is expected to be operational on January 1st/ 2015. The objective of this visit was to verify the Colombian municipalities involved in the project development as well as the impact of the project in this lands, in order to check its management in the environmental and social plans implemented by the PP.

2.5 Resolution of Any Material Discrepancy

The validation audit process was prepared following the site visit during which all issues identified during the desk review were discussed. Corrective action and clarification requests raised by ICONTEC were presented to the PP and resolved through communication and meetings between Central Hidroeléctrica El Edén S.A.S E.S.P and ICONTEC. To guarantee the transparency of the validation process, the concerns raised and the response provided by the project participants are documented in more detail in Annex A.

3 VALIDATION FINDINGS

3.1 Project Design

ICONTEC had the opportunity to verify the following description in the on-site visit to the project proponent:

Title of project activity:	Hydroelectric Project El Edén
Project Proponent:	Central Hidroeléctrica El Edén S.A.S E.S.P Cll. 18 Av. Las Palmas Nro. 35-69 Centro Empresarial Palms Avenue, Of. 424, Medellín – Colombia.
Project Start Date:	January 1 st 2015
Project's crediting period:	10 years.
Project Scale:	Project
Estimated amount of GHG emission reductions over the chosen crediting period :	321,024 tonnes CO ₂ e
Location of the project activity:	The project area includes the following municipalities: Pensilvania, Marquetalia and Manzanares in the department of Antioquia, Colombia. The powerhouse is located in the following coordinates (latitude/longitude): E 885038 N 1079832 (datum Bogotá)
Methodology :	ACM0002, Consolidated baseline methodology for grid-connected electricity generation from renewable sources, version 13.0.0./26/ This methodology is used in combination with the “Tool to calculate the emission factor for an electricity system” ver. 03.0.0 /27/ and “Tool for the demonstration

and assessment of additionality” ver. 07.0.0 /28/

Project scope, type, technologies and measures implemented, and eligibility of the project:

The Project consists of a new run-of-river hydroelectric power plant to take advantage of the waters of the la Miel River. The Project will comprise a powerhouse with 2 x 10.115 MW Pelton turbines/6/ (page 10). The Project will be connected Colombian Interconnected electric system by a 115 kV transmission lines. In the version 01 of PD the documental evidence regarding to the annual energy generation was not provided, hence ICONTEC raised CAR 1 and the project proponent provided this evidence and a new version of PD (version 02) with a mention about annual average energy production (104.5 GWh) supported by documental reference /5/.

The GHG project assessed can be classified as a VCS project in the sector 1, Energy (renewable/non-renewable), according to the List of Sectoral Scopes of VCS.

The project is expected to be operational on January 1st/2015 and this date was defined as the starting date of the crediting period.

Project compliance with applicable laws, statutes and other regulatory frameworks

As described in Section 1.11 of the PD (version 04), the Project is in compliance with applicable Colombian laws. In order to validate this statement the following evidence was reviewed: /7/, /8/, /9/, /10/, /11/, /12/ and /13/.

Ownership and Right of use

ICONTEC reviewed documental evidence /8//14/, in order to validate the unconditional, undisputed and unencumbered Central Hidroeléctrica El Edén S.A.S E.S.P’s ability to claim the GHG emission reduction that Hydroelectric Project El Edén will generate or cause, since these documents describe an enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant and its equipment (Central Hidroeléctrica El Edén S.A.S E.S.P). ICONTEC considered these documental evidences as credible and reliable.

Project does not apply to other emission trading programs or other forms of environmental credit and is not part of any compliance scheme (binding limits).

ICONTEC confirms that PD version 04 depicts in an accurate and complete way the project description, and it complies with the relevant form and guidance⁷.

⁷ Project Description Template, version 3.1, issued on October 4th, 2012

3.2 Application of Methodology

3.2.1 Title and Reference

The approved UNFCCC baseline and monitoring methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version 13.0.0) /26/ has been applied by the project activity.

3.2.2 Applicability

The project complies with the applicability criteria of the methodology /26/ as ICONTEC verified, as follows:

Methodology for applicability conditions analysis

Applicability condition	Means of validation
This methodology is applicable to grid-connected renewable power generation project activities that: a) Install a new power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant). b) Involve a capacity addition. c) Involve a retrofit of (an) existing plant(s). d) Involve a replacement of (an) existing plant(s).	Hydroelectric Project El Edén is the installation of a new hydro power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant). ICONTEC verified this statement by means of: - On site visit - Documental Review of : • Technical Description about hydroelectric use of la Miel River – Project El Edén /4/. • Technical offer for electromechanical equipment of El Edén hydroelectric project /6/
The project activity is the installation, capacity addition, retrofit or replacement of a power plant/unit of one of the following types: hydro power plant/unit (either with a run-of-river reservoir or an accumulation reservoir), wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit	Hydroelectric Project El Edén is the installation of a run-of-river hydro power plant. ICONTEC verified this statement by means of: - On site visit - Documental Review of : • Technical Description about hydroelectric use of la Miel River – Project El Edén /4/. • Technical offer for electromechanical equipment of El Edén hydroelectric project /6/

	Also, they were reviewed next documents from official sources: Letter issued by UPME acknowledging the registration of the El Edén Hydroelectric Project as a future generation project to be connected to Colombian Electrical grid/15/.
In the case of capacity additions, retrofits or replacements (except for wind, solar, wave or tidal power capacity addition projects which use Option 2: on page 10 /26/ to calculate the parameter $EG_{PJ,y}$): the existing plant started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion or retrofit of the plant has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	According to the documental review carried out by ICONTEC /4/,/6/, Hydroelectric Project Ituango will be a Greenfield hydroelectric power plant and this applicability condition does not apply.
In case of hydro power plants one of the following conditions must apply: • The project activity is implemented in an existing single or multiple reservoirs, with no change in the volume of any of reservoirs; or • The project activity is implemented in an existing single or multiple reservoirs, where the	Hydroelectric Project El Edén will be a run-of-river power plant hence there is not a reservoir, therefore there is no change in the volume of any reservoirs. ICONTEC verified this statement by means of : Documental review of : • Technical Description about hydroelectric use of la Miel River – Project El Edén /4/. • Environmental Impact Study about hydroelectric use of la Miel River – Project El Edén /7/

volume of any of reservoirs is increased and the power density of each reservoir, as per the definitions given in the project emissions section, is greater than 4 W/m^2; or • The project activity results in new single or multiple reservoirs and the power density of each reservoir, as per the definitions given in the project emissions section, is greater than 4 W/m^2.	
In case of hydro power plants using multiple reservoirs where the power density of any of the reservoirs is lower than 4 W/m^2 all the following conditions must apply: • The power density calculated for the entire project activity using equation 5 is greater than 4 W/m^2; • Multiple reservoirs and hydro power plants located at the same river and where are designed together to function as an integrated project that collectively constitute the generation capacity of the combined power plant; • Water flow between multiple reservoirs is not used by any other hydropower unit which is not a part of the project activity; • Total installed capacity of the power units, which are	These applicability conditions are not applicable to the project activity, as it was explained above

driven using water from the reservoirs with power density lower than 4 W/m², is lower than 15 MW; • Total installed capacity of the power units, which are driven using water from reservoirs with power density lower than 4 W/m², is less than 10% of the total installed capacity of the project activity from multiple reservoirs.	
The methodology is not applicable to the following: • Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; • Biomass fired power plants; • A hydro power plant that results in the creation of a new single reservoir or in the increase in an existing single reservoir where the power density of the power plant is less than 4 W/m².	Hydroelectric Project El Edén will be a run-of-river power plant hence there is not a reservoir. ICONTEC verified this statement by means of : Documental review of : • Technical Description about hydroelectric use of la Miel River – Project El Edén /4/.
In the case of retrofits, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, i.e. to use the power generation equipment that was already in use prior to the	According to the nature of the project (greenfield project), this condition is not applicable to Hydroelectric Project El Edén

implementation of the project activity and undertaking business as usual maintenance.	
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In summary, the proposed project activity meets the applicability conditions of ACM0002 (version 13.0.0).

3.2.3 Project Boundary

In accordance with the applied methodology /26/, the project boundary is “The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to”. This statement was verified by ICONTEC by means of documental review of technical description /4/ and the Letter issued by UPME acknowledges the registration of the El Edén Hidroelectric Project as a future generation project to be connected to Colombian National Interconnected System /15/. The coordinates of the power house are correctly documented in the VCS PD. The sources and sinks of greenhouse gas identified in the VCS PD are deemed to be appropriate.

	GHG involved	Means of Validation
Baseline emissions	CO ₂	Emissions from the generation of electrical power by fossil power plants in Colombian National Power Grid (Section 3.1 and 3.2.2 in this report)
Project emissions	CH ₄	The project activity has no reservoir. Therefore, project emissions are neglected. (Section 3.2.2 in this report)
Leakage	-	Considered to be neglected as per ACM0002 (version 13.0.0) /26/

In accordance with the project activity and the applied methodology /26/, the emission sources are properly described in the PD version 04. The greenhouse gas emissions occurring within the project boundary as a result of its implementation are all addressed by the applied methodology, therefore there are not greenhouse gas emissions within the project boundary caused by the implementation of the project activity which contribute to more than 1% of the expected annual emission reductions and which are not addressed in by the applied methodology. This was verified by ICONTEC by means of the documental review related with the project's works and its nature/4//7/.

3.2.4 Baseline Scenario

The project activity comprises the installation of a new grid-connected renewable power plant. Consequently, according to the applied methodology, the baseline scenario is as following: *“Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as*

reflected in the combined margin (CM) calculations described in the Tool to calculate the emission factor for an electricity system.”

The PD version 04 correctly identifies the baseline scenario as presented above. The relevant grid is the Colombian National Interconnected Electricity System (SIN)./15/

As methodology ACM0002 (version 13.0.0) prescribes the baseline scenario and no further analysis is required, there is no need to take steps to identify the baseline scenarios.

3.2.5 Additionality

To demonstrate the additionality of the Project, the PD has correctly applied the “Tool for the demonstration and assessment of additionality” (Version 07.0.0) /28/. PP used an investment analysis to determine that the project is additional. No Barrier Analysis was presented. The details of ICONTEC’s assessment on the project additionality are described below:

Step 0: Demonstration whether the proposed project activity is the first-of-its-kind

Since in the Tool for the demonstration and assessment of additionality /28/ clause 14 states that this step is optional and the PP did not use this analysis, hence it is considered that the proposed project activity is not the first-of-its-kind.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

In the Tool for the demonstration and assessment of additionality /28/ clause 8 states that for project activities that apply this tool in context of approved consolidated methodology ACM0002, only need to identify that there is at least one credible and feasible alternative that would be more attractive than the proposed project activity; however the PP has presented in the PD version 04 two alternatives:

Analysis of alternatives

Alternatives	Assessment	Means of Validation of Alternatives	Conclusion
1. Undertake the Hydroelectric Project El Edén as an hydroelectric power generation without being registered as a VCS (CDM) project activity	In accordance with the tool for demonstration and assessment of additionality /28/ clause 20 (a)	All the alternatives comply with all applicable and enforced Colombian legislation. Besides that, ICONTEC could confirmed by means of the lead auditor’s knowledge of the sector and documental review of	- The Identification of alternatives is realistic and credible. - The alternatives are consistent with
2. Not undertake the	It corresponds to current		

proposed project activity	situation, that is, the electricity will continue to be generated by the existing generation mix operating in the grid and by the addition of new generating sources. Besides, this alternative is the baseline prescribed in the applied methodology /26/	the Colombian regulatory framework ⁸ that alternatives are credible and realistic in the Colombian scenario	mandatory laws and regulations
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Step 2: Investment Analysis

The additionality was demonstrated according to Tool for the demonstration and assessment of additionality /28/ and the Guidelines on the assessment of investment analysis /29/.

For this project activity, benchmark analysis (Option III) was selected for the investment analysis. In accordance with the type of this project activity it is expected that incomes other than VCS (CDM) inputs will be generated, regarding energy sales to the market, so ICONTEC agreed that simple cost analysis is not suitable. On the other hand, taking into account that Central Hidroeléctrica El Edén S.A.S E.S.P has no intention in building a power station, different to hydroelectric in the area of the project activity as demonstrated in the PD and related documentation /4//5//6//7/, an investment comparison analysis is not appropriate and a benchmark analysis was used to determine the profitability of the investment.

The benchmark was chosen in accordance with the Appendix in the Guidelines on the assessment of investment analysis /29/ which provides default values for the expected return on equity for different project activities, the relevant benchmark for energy projects in Colombia (Group 1 as given in the guidelines /29/) is 12%, however these values are expressed in real terms, since the investment analysis carried out by the PP was performed in nominal terms, the addition of the Colombian inflation rate was used to convert the real term value (12%) to nominal term.

ICONTEC reviewed the Colombian Central Bank's Website⁹ and did not find neither inflation forecast nor target inflation rate for the duration of the crediting period, hence the PP correctly chose the average forecasted inflation rate for the next five years published by the IMF¹⁰ (International Monetary Fund World Economic Outlook), in accordance with the requirement stated in the guidelines /29/ clause 7. This average forecasted inflation is 2.901% but in order to work the rate along the financial calculation with only 1 decimal, this value was rounded to 2.9%.

⁸ <http://www.creg.gov.co/cxc/secciones/documentos/leyes.htm>

⁹ <http://www.banrep.gov.co/es/contenidos/inflacion-basica>

¹⁰ <http://www.imf.org/external/pubs/ft/weo/2011/02/weodata/weorept.aspx?pr.x=83&pr.y=6&sy=2010&ey=2016&scsm=1&ssd=1&sort=country&ds=.&br=1&c=233&s=PCPIPCH&grp=0&a>

Therefore the benchmark chosen by the PP was 14.9% (12% + 2.9%) as Nominal Return on equity in order to be compared with the financial indicator most suitable for the project type and decision context, in this particular case is IRR. ICONTEC agrees with all the data used in benchmark determination and points out that they were clearly presented, available to consult and correct.

ICONTEC reviewed the file “Equity IRR Hydroelectric Project El Edén (28Aug13).xls” and confirmed that it included the relevant cost and revenues. The period of assessment was 33 years, in accordance with the Clarification to the Guideline /29/¹¹ which establishes: *If project participants choose a renewable crediting period and if the technical lifetime of the CDM project activity is more than 20 years, the investment analysis shall be conducted for 20 years and include the fair value of the project activity assets at the end of the assessment period.* Since the renewable crediting period in VCS framework is 10 years and on the other hand the operation stage (50 years) and the construction stage (30 months) /8//17/ are more than 20 years, the audit team deemed appropriate to undertake the investment analysis for 33 years. The fair value of the project activity assets was included at the end of the assessment period, see cell AJ 53, sheet “Equity IRR” of the file Equity IRR Hydroelectric Project El Edén (28Aug13).xls

The input values used in all investment analysis have been consistently applied at the time of the investment decision (May 16th, 2012) /17/ and in all calculations. The depreciation has been added back to the net profits, this item has been made in accordance with Colombian accounting standards and they are aligned with Guidelines on the assessment of investment analysis/29/, in addition, taking into account the Guidelines on the assessment of investment analysis /29/ clause 18, the PP has assumed as a default a 50% debt / 50% equity finance structure, the debt terms assumptions as: grace period (30 months = construction period), term (10 years), interest rate (11.20%) and annual principal (USD\$ 2,414,215 = Total Debt/Debt Term) are based on parameter that are standard in the Colombian market.

The following table shows how ICONTEC assessed the main parameters used in the cash flow:

Investment Analysis Parameters

Parameter	Value	Source	Validation Analysis
Energy generation for sales	104.5 GWh/year	Hydrologic Studies carried out by I-Consult to El Edén hydroelectric project /5/	In assessing whether the energy generation has been defined correctly, ICONTEC used the Guidelines for the reporting and validation of plant load factors /30/, and verified that the third-party generation study /5/ described by the PP as the documental source of this value, was provided by I-Consult S.A. ¹² .
Firm Energy	43.5 GWh/year		
Surplus Energy	61 GWh/year ¹³		

¹¹ See https://cdm.unfccc.int/filestorage/n/v/extfile-20130604103656299-meth_guid53.pdf/meth_guid53.pdf?t=R0l8bXQwb2dvfDDSgrz3xg-mpy_TXcZHxegi

¹² <http://iconsult.com.co/home/index.html>

¹³ 61 GWh/year = 104.5 GWh/year – 43.5 GWh/year

			The hydrologic studies/5/ determined a plant load factor of 61.5%, as well as the firm energy based on model for turbinated water volume In accordance with the clause 3 (a) of the mentioned Guidelines /30/, ICONTEC deemed this value as credible and reliable.
Energy tariff I (up to 42.4 GWh/year)	COP\$132/kWh		In order to assess the reliability of these values, ICONTEC reviewed the report about tariffs and commercial transactions on the Colombian Wholesale Market/18/, and noticed that the average tariff for regulated market was COP\$131.05/kWh and the average tariff for non-regulated market was COP\$102.79/kWh, therefore these values used by the PP in its cash flow assumption were considered by ICONTEC as credible.
Energy tariff II (surplus energy)	COP\$105/kWh	Based on proposal for energy sales available at decision making /16/	In accordance with the energy purchase proposal /16/, these tariffs should be increased by the PPI (Colombian Producer Price Index). The PP had into account this consideration and included a worksheet named <i>PPI</i> in the spreadsheet "Equity IRR Hydroelectric Project El Edén (28Aug13).xls" which contains the average value for Colombian Producer Price Index from 2009 to 2011 downloaded directly from DANE¹⁴ Webpage¹⁵.
Total Investment	USD\$ 48,284,307	Construction Contract for the infrastructure implementation and operation of El Eden Hydroelectric project /14/	ICONTEC reviewed the Construction Contract signed between Central Hidroeléctrica El Edén S.A.S E.S.P and UT-GC CHOC /14/ in order to verify the investment costs reported in the cash flow file /3/. The price per kW installed is US\$2,386.77. This unit cost was cross checked by ICONTEC with the one reported in "Renewable Power Generation Costs in 2012, An Overview"/19/, issued by IRENA, page 44, figure 5.4, where is described that "The total installed costs for large-scale hydropower projects in Latin America typically range from a low of USD 1,000/kW to around USD 3,500/kW", hence the Investment cost reported for El Edén hydroelectric power plant is inside the recognized worldwide range of investment costs reported by IRENA, therefore ICONTEC deems that estimations of investment in the power

¹⁴ Colombian National Administrative Department of Statistics (in Spanish: Departamento Administrativo Nacional de Estadística)

¹⁵ <http://www.dane.gov.co/index.php/indices-de-precios-y-costos/indice-de-precios-al-productor-ipp> Data serie: "Índice de Precios del Productor (IPP), según clasificación CUODE / 2001 - 2013 (abril)"

			electricity plant are suitable, reliable and conservative. In accordance with the construction contract /14/, this value should be increased by the CPI (Colombian Consumer Price Index). The PP had into account this consideration and included a worksheet named <i>CPI</i> in the spreadsheet “Equity IRR Hydroelectric Project El Edén (28Aug13).xls” which contains the average value for Colombian Producer Price Index from 2009 to 2011 downloaded directly from DANE Webpage¹⁶.
Expenses (this includes operation, maintenance, administration and commercial costs)	US\$ 641,070	This data is provided by the project participant in accordance with its previous experience /24/	In order to carry out the crosscheck, ICONTEC deems that as a common practice, O&M costs are usually given as a percentage of investment cost, so this ratio is: $\frac{O\&M\ Costs}{Investments\ Costs} = \frac{US\$ 641,070}{US\$ 48,284,307} = 0.0133$ ICONTEC could verify that the O&M costs reported by PP (1.33% of the Investments Costs) are lower than the ones reported in “IPCC, 2011: IPCC Special Report on Renewable Energy Sources and Climate Change Mitigation. Prepared by Working Group III of the Intergovernmental Panel on Climate Change /25/, page 484, Table 5.8”, where the O&M costs are the 2.5% of the total Investment cost, ICONTEC deems the value used as conservative, credible and reliable
Income tax	33 %	Tax Law 1111/06 (Ley de Impuesto sobre la Renta). The applicable rate is 33% as per article 12 in the law	With the aim to assess the value described by the PP, ICONTEC verified the following link: http://www.secretariassenado.gov.co/senado/basedoc/ley/2006/ley_1111_2006.html, clause 383. The truthfulness of the information provided by the PP was checked.
Equipment cost (for depreciation only)	USD\$ 3,954,333	Quotation made by Andritz Hydro for electromechanical equipment for El Edén Hydroelectric project /20/	The depreciation costs are based on the rules established in the Decree 3019/1989 /23/. The individual investment costs with different depreciation rates (20 years straight-line for civil works and 10 years straight-line for equipment) were not available at decision making. Therefore the equipment costs as per the proposal /20/ were used for equipment costs. Since the faster depreciation of equipment

¹⁶ http://www.dane.gov.co/index.php?Itemid=76&id=103&option=com_content&view=article Data serie: Variaciones Porcentuales IPC / 1998 - 2013 (abril)

Annual depreciation Civil Works + Infrastructure	20 years	In accordance with Colombian accounting standards	leads to a slightly lower tax rate during the project beginning, the IRR tends to increase slightly and this assumption was deemed as conservative by ICONTEC.
Annual depreciation equipment	10 years		

ICONTEC could confirm that the assumptions used are appropriate and the financial calculations made in the file “Equity IRR Hydroelectric Project El Edén (28Aug13).xls” are correct.

With a benchmark of 14.9%, the financial analysis for this project activity states that the Equity IRR of the project without VCS revenues is 12.22%.

Regarding to sensitivity analysis, the PP has correctly chosen the Investment cost, the Electricity tariff, the energy generation and the O&M costs as variables for this purpose, with a variation of $\pm 10\%$. The outcome of this sensitivity analysis is described in Table 7 of PD version 04. As can be seen, in all of scenarios the project IRR is less than benchmark IRR. Therefore, in the view of ICONTEC, the sensitivity analysis demonstrates that the project is highly probable not to be financially attractive without the VCS incomes.

Step 3: Barrier Analysis

No Barrier analysis is presented in the PD version 04.

Step 4: Common Practice Analysis

Step 4 of the additionality analysis in the PD version 04 has been prepared according to section 4.5.1 of the latest version of the Tool for demonstration and assessment of additionality /28/, which applies the Guideline on Common Practice /31/

The geographical scope for the common practice analysis is Colombia. Considering that the project is connected to the Colombian Electric Interconnected System, ICONTEC confirmed that it is justified to select Colombia as the geographical scope in accordance with the Guidance /31/.

The project has an installed capacity of 20.230 MW. Therefore, the PD version 04 correctly defines the applicable range as 10.115 MW – 30.345 MW. ($\pm 50\%$ of the installed capacity of the project activity.)

The additionality tool defines output as “good/services produced by the project activity including, among other things, heat steam, electricity, methane, and biogas unless otherwise specified in the applied methodology.” Therefore, in the case of this project, the output considered is the electricity generated by grid-connected power plants.

According to the PD version 04 and the spreadsheet used to common practice analysis /21/, N_{all} is 23. ICONTEC verified this information by means of accessing the XM’s service information

(named NEON)¹⁷, and also the UNFCCC Website and VCS Website in order to verify if the registered CDM project activities, project activities submitted for registration and project activities undergoing validation were excluded. ICONTEC verified that the information provided by the PP regarding N_{all} is traceable, reliable and credible.

In accordance with the Guidelines on common practice /31/ clause 4) d (Different technologies), the PP correctly argued Investment Climate, since in 1994 the laws 142 and 143 were enacted, these two laws define the ongoing electrical market structure and regulatory framework¹⁸ and changed completely the control and development of the electric sector in Colombia. ICONTEC carried out documental review /22/ in order to validate the date on which the projects listed in N_{all} estimation started commercial operation, the energy source/fuel and the size of the power plants. Therefore, ICONTEC could confirmed the estimation of $N_{diff} = 22$ presented by the PP in the PD version 04 is traceable, reliable and credible, and it is aligned with the Guidelines' requirements.

$$\text{Hence: } N_{all} - N_{diff} = 23 - 22 = 1; \text{ and } F = 1 - \frac{N_{diff}}{N_{all}} = 1 - \frac{22}{23} = 0.0435$$

Consequently, Hydroelectric Project El Edén is not a common practice in Colombia since F is lower than 0.2 and $N_{all} - N_{diff}$ is lower than 3.

Additionality Conclusion

In conclusion, it was verified that the project is not the most likely baseline scenario. Hence, the emission reductions occurring from the project are deemed additional to those that would occur in the absence of the project activity.

3.2.6 Quantification of GHG Emission Reductions and Removals

The steps taken, the equations and parameters applied in the VCS PD version 04 to calculate baseline emissions, project emissions, leakage and emission reductions comply with the requirements of the selected methodology/26/ including applicable tool(s)/27/. Please see below:

Baseline emissions need to be calculated in accordance with equation (6) of the relevant methodology /26/:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS (CDM) project activity in year y (MWh/yr)

¹⁷ <http://sv04.xm.com.co/neonweb/>

¹⁸ <http://www.creg.gov.co/cxc/secciones/documentos/leyes.htm>

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh) /27/

If the project activity is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, then:

$$EG_{PJ,y} = EG_{facility,y}$$

Where

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS (CDM) project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

In the CERs calculation spread sheet (EF Colombia 2010-2012.xls)/2/ and in the PD version 04, PP took $EG_{facility,y}$ as the expected net electricity generation supplied by the project plant to the Colombian grid in year y (MWh/yr): 104,500 MWh/year.

The baseline emission factor ($EF_{grid,CM,y}$) was calculated by the PP using the Tool to calculate the emission factor for an electricity system /27/ as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) factors, according to the following steps:

Step 1. Identify the relevant electricity systems:

For Hydroelectric Project El Edén case, the PP used the Colombian transmission system as the project electricity system, which is operated and maintained by XM¹⁹. ICONTEC agreed taking into account that Hydroelectric Project El Edén will be connected to this system /13/.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional).

In step 2, the PP chose option I, only grid power plants are included in the calculation.

Step 3. Select a method to determine the operating margin (OM).

In the step 3, simple adjusted OM was chosen by the PP to calculate the operating margin emission factor, using ex-ante data vintage taking into account that in the Colombian electric system, the low-cost/must-run resources constitute more than 50% of total grid generation, hence

¹⁹ XM is also in charge of the National Dispatch Center (CND).

the Simple OM method cannot be used as it was verified by ICONTEC by means of reviewing the generation average of the five most recent years²⁰. ICONTEC validated this choice.

Step 4. Calculate the operating margin emission factor according to the selected method.

Calculations of OM emission factor (step 4) were made as illustrated in the PD version 04, section 3.1, which is according to the tool specifications/27/. Plant emission factors $EF_{EL,m,y}$, were calculated under option A1 because the data on fuel consumption and energy generation are available. ICONTEC validated the values comparing the values presented for the PP in the Emissions reduction file/2/, against the values downloaded from the XM website (ICONTEC made a random sampling of 4 plants in four different dates and hours).

After the comparison, ICONTEC deemed reliable and appropriate the values used. The OM calculated was 0.4798 tCO₂/MWh, hence ICONTEC deemed the obtained value as reliable and credible.

Step 5. Calculate the build margin (BM) emission factor.

In order to calculate the BM emission factor (step 5) option 1 (ex-ante) for the first crediting period was adopted, the build margin emission factor is calculated based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. The PP took the information from the latest official XM statistics.

ICONTEC agreed with the data collection used for the PP to calculate the BM, hence the BM is confirmed as reliable and credible. The BM calculated was 0.1346 tCO₂/MWh, hence ICONTEC deemed the obtained value as reliable and credible

Step 6. Calculate the combined margin (CM) emission factor.

Finally, combined margin was correctly calculated by weighted average method, as it is explained below:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where:

$$EF_{grid,OM,y} = 0.4798 \text{ tCO}_2/\text{MWh}$$

$$EF_{grid,BM,y} = 0.1346 \text{ tCO}_2/\text{MWh}$$

$$W_{OM} = 0.5$$

$$W_{BM} = 0.5$$

Obtaining a result for the $EF_{grid,CM,y} = 0.3072 \text{ tCO}_2/\text{MWh}$

In conclusion, ICONTEC confirmed that the value included in the PD (version 04) for BE_y (321,024 tCO₂e) has been justified adequately.

²⁰ <http://sv04.xm.com.co/neonweb/>

Hence ICONTEC deemed the obtained value by the PP as reliable.

According to the previous description ICONTEC found that the project participant has correctly applied the selected methodology/26/ with respect to the Baseline determination. All estimates of the baseline emissions can be replicated using the data and parameter values provided in the PD version 04.

Regarding to project emissions, these emissions need to be calculated in accordance with equation (1) of the relevant methodology/26/:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y = Project emissions in year y (tCO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO₂/yr)

$PE_{GP,y}$ = Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr).

According to ACM0002 v13.0.0, the only possible source of project emissions for hydro power plants are emissions from reservoir ($PE_{HP,y}$). These emissions from reservoir are calculated in accordance to the power density of the single or multiple reservoirs.

Since the project activity has no reservoir, this calculation is not required; therefore, in accordance with the applied methodology /26/ $PE_{HP,y} = 0$.

Regarding to the leakage, and according to ACM0002 v13.0.0, no leakage emissions need to be considered. The PD version 04 correctly describes this assumption.

Emission reductions are calculated in accordance with equation (11) of the relevant methodology/26/:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emissions in year y (tCO₂/yr)

PE_y = Project emissions in year y (tCO₂e/yr)

Seeing that project emissions is zero, $ER_y = BE_y$. See above, section 3.4 on this report how ICONTEC was able to validate the BE_y values presented in the PD version 04.

In conclusion, ICONTEC confirmed that the value included in the PD (version 04) for ER_y (321, 024 tCO₂e) has been justified adequately.

3.2.7 Methodology Deviations

The project activity's implementation is fully in accordance with ACM0002 "Consolidated baseline methodology for grid-connected electricity generation from renewable sources" version 13.0.0 /26/ and hence deviation of methodology is not applicable.

3.2.8 Monitoring Plan

ICONTEC hereby confirms that the monitoring plan complies with the requirements of the methodology/26/ and the applicable tools/27/.

The Project uses the methodology ACM0002 Consolidated baseline methodology for grid-connected electricity generation from renewable sources, version 13.0.0 /26/. The project involves the installation of a new grid connected hydro power plant.

The steps taken to assess whether the monitoring arrangements described in the monitoring plan are feasible within the project design are described below:

Data and Parameters Available at Validation:

In PD version 04, the PP has fixed the Combined margin CO₂ emission factor for the Colombian Electrical grid, its value: 0.3027 tCO₂/MWh as it was described in section 3.2.6 on this report

Data and Parameters Monitored:

EG_{facility,y} = In accordance with the monitoring plan, the main parameter that will be monitored is the quantity of net electricity generation supplied by the project plant to the grid in year y, measured continuously by the power plant's meters installed in the Project's substation (connection point to the grid). The measurement will be continuously done and recorded monthly.

Management system and quality assurance

The monitoring arrangement for the project has been described in the VCS PD:

- The accuracy of electric meters (0.2S) and record frequency have been identified appropriately.
- The operation and management structure of the monitoring have been described appropriately.
- The electric meters will be calibrated according to the Colombian national standard NTC 4856.
- The emergency procedure and the correctness action have been identified appropriately.

In summary, the validation team is convinced that the monitoring plan reflects good monitoring practice appropriate to the project type and the PP is able to implement the monitoring plan.

3.3 Environmental Impact

According to the legislation in force (law 99 from 1993²¹), an Environmental Impact Study (EIA) is required to be carried out before the project can receive its environmental license/8/.

The project participants have undertaken an analysis of environmental impacts and an environmental impact assessment was prepared in accordance with procedures as required by Colombian legislation. The EIA /7//9/, identified the project's impacts on the environment and also the necessary mitigation actions.

During the visit to the project proponents, they explained that in accordance with the Colombian legislation, it is necessary to perform an Environmental accomplishment Report periodically (ICA by its Spanish Acronym). ICONTEC verified that these reports /10//11//12/²² are made with the aim to report to the CORPOCALDAS the compliance in the obligations stated in the Environmental License granted.

ICONTEC based on the above arguments validated that the project is carrying out the environmental mitigation actions (environmental programs) in accordance with national legislation.

3.4 Comments by stakeholders

The validation team assessed the adequacy of the local stakeholder consultation as follows:

Identification of local stakeholders

The local stakeholders involved in the proposed project activity have been appropriately identified. They are the local authorities (CORPOCALDAS²³) and affected local residents.

Timing of stakeholder consultation process

The local stakeholder consultation was processed in three rounds, the first one from August 2010 to May 2011, the second one from June to August 2011, and third one from March to May 2012.

²¹ http://www.secretariassenado.gov.co/senado/basedoc/ley/1993/ley_0099_1993.html

²² The project should present periodically to the environmental authority an ICA report, where is explained the accomplishment of the commitments established in the environmental license.

²³ Regional Autonomous Corporation of Caldas (In Spanish: Corporación Autónoma Regional de Caldas)

Summary of comments provided

By means of reviewing local stakeholder consultation meeting record and interview with local stakeholders, the validation team confirms that summary of comments provided has been addressed in the VCS PD appropriately and transparently.

4 VALIDATION CONCLUSION

ICONTEC performed the validation of the Hydroelectric Project El Edén, in Colombia, the validation was performed on the basis of Verified Carbon Standard (VCS version 3.3) and all relevant VCS requirements, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The review of the Project Description and the subsequent follow up interviews has provided ICONTEC sufficient evidence to determine the fulfilment of the stated criteria.

The project activity is being proposed by Central Hidroeléctrica El Edén S.A.S E.S.P. and correctly applies the methodology: ACM0002: Consolidated baseline methodology for grid-connected electricity generation from renewable sources, Version 13.0.0

The project consists on the development of a new hydroelectric power plant that takes advantage of the waters of the Cauca River. The water flow will allow an installed turbine capacity of 20.230 MW and annual supply of electricity to the grid of 104.5 GWh. It is demonstrated that the project is not a likely baseline scenario. Emission reductions attributable to the project are hence additional to any that would occur in the absence of the project activity.

The total emission reductions from the project are estimated to be 32,102 tCO₂e per year over the selected 10 years crediting period. The emission reduction forecast has been checked and it is deemed likely that the stated amount is achieved because the underlying assumptions do not change.

In summary, it is ICONTEC's opinion that the Hydroelectric Project El Edén in Colombia, as described in the PD version 04, meets all relevant VCS requirements and correctly applies the baseline and monitoring methodology ACM0002: Consolidated baseline methodology for grid-connected electricity generation from renewable sources, Version 13.0.0. ICONTEC thus requests the registration of the project as a VCS project activity.

5 ANNEX A. RESOLUTION OF CORRECTIVE ACTION, FORWARD ACTION AND CLARIFICATION REQUEST

Report clarifications and corrective action requests	Reference	Summary of project owner response	Validation conclusion
CAR 1 <i>The documental evidence regarding to the annual energy generation used for emissions reduction calculations and additionality assessment is not provided</i>	Guidelines for the reporting and validation of plant load factors, version 01.0, section II	<i>See attached energy generation assessment from I-Consult " Análisis horario El Edén de energía DEF.xls"</i> <i>The values were adjusted in the PD and a brief explanation with the correct reference was added to the technical description and additionality sections.</i> Further answer: <i>We attach the email sent from I-Consult to El Edén's responsible with the energy generation analysis. This demonstrates that the analysis was made and sent by I-Consult. We also consider that the presentation of I-Consult to Icontec during the validation visit gives further credibility to this evidence, since the analysis were discussed and presented directly by I-Consult.</i>	Validation team response: <i>It is not possible to trace if the document provided by the PP " Análisis horario El Edén de energía DEF.xls" was issued by a third party (I-Consult), or if this data was provided to banks and/or equity financiers while applying the project activity for project financing, or to the government while applying the project activity for implementation approval.</i> <i>In a further answer to this finding the PP provided an evidence that demonstrate that energy production estimations were made by a third party (I-Consult)</i> Validation team conclusion: Closed

CAR 2 The $EF_{grid,OM-adj,y}$ calculations are not based on the most recent data available.	Tool to calculate the emission factor for an electricity system, version 03.0.0, clause 36, (a)	The emissions factor has been updated based on the most recent data available from the electricity grid administrator XM. The updated EF uses the vintage years 2010, 2011 and 2012. All calculations have been adjusted.	Validation team response: The lead auditor verified the data provided by the PP in the file "EF Colombia 2010-2012.xls" against the publicly official and available source, and confirmed that for $EF_{grid,OM-adj,y}$ calculations are based on the most recent data available Validation team conclusion: Closed
CAR 3 The installed capacity value used for common practice assessment is not consistent with the project installed capacity	Guidelines on common practice, version 02.0, clause 5	The installed capacity of 20.230 MW has been used for the updated common practice analysis. See updated PD and Excel File "Common Practice Analysis El Edén"	Validation team response: The value used for common practice assessment is 20.23 MW, this value is consistent with the installed capacity determined for the project

			activity Validation team conclusion: Closed
CAR 4 The identification of N_{all} in the common practice analysis was not taken into account all the registered CDM project activities, or project activities submitted for registration, or project activities undergoing validation	Guidelines on common practice, version 02.0, clause 7	This has been corrected. See updated PD and Excel File "Common Practice Analysis El Edén"	Validation team response: The identification of N_{all} has been determined using the latest version of the Guidelines on common practice issued by the Executive Board of CDM, hence it was taken into account all the registered CDM project activities, or project activities submitted for registration, or project activities undergoing validation Validation team conclusion: Closed
CAR 5 The arguments provided by the PP in order to identify those power plants that	Guidelines on common practice, version 02.0, clause 4 (e)	This has been corrected. See updated PD and Excel File "Common Practice Analysis El Edén". All detailed references are provided in the Excel. Basically, the power plants in the same capacity range (+/- 50%) are either	Validation team response: The PP has provided to the lead auditor the documental

<i>apply technologies that are different to the technology applied in the proposed project activity (more specifically: "Other features") are not traceable by means of documental evidence</i>		<i>thermal power plants or were constructed before the new energy law in 1994, which means they had a very different regulatory framework and were developed with resources from the state.</i>	<i>evidence to identify those power plants that apply technologies that are different to the technology applied in the proposed project activity in accordance with the latest version of the Guidelines on common practice issued by the Executive Board of CDM.</i> <i>Validation team conclusion:</i> <i>Closed</i>
CAR 6 <i>The energy tariff parameter was not included in the sensitivity analysis</i>	<i>Guidelines on the assessment of investment analysis, version 05.0, clause 20</i>	<i>Corrected. See updated financial analysis.</i> Further answer: <i>See "Equity IRR Hydroelectric Project El Edén (11Jun13)" with the updated values. The version attached to the last validation protocol was not the correct one. Please use this one.</i>	<i>Validation team response:</i> <i>In the PD version 02 (page 21) it was included a summary of sensitivity analysis made for energy tariff parameter, however the updated version of the spreadsheet with the financial analysis used for additionality assessment was not provided to the lead auditor.</i> <i>In a further answer to this finding the PP provided the updated version of the spreadsheet with the financial</i>

			<i>analysis used for additionality assessment</i> <i>Validation team conclusion:</i> <i>Closed</i>
CL 1 <i>The project description was not developed in the latest version of the VCS project description template</i>	<i>VCS project description template, version 3.1</i>	<i>The newest version 3.1 has been used for the updated PD.</i>	<i>Validation team response:</i> <i>The updated project description (PD version 02) was developed in the latest version of the VCS project description template</i> <i>Validation team conclusion:</i> <i>Closed</i>
CL 2 <i>In section 6 of the PD version 01 was not provided the information regarding the local stakeholder consultation undertaken by the PP.</i>	<i>VCS project description template, version 3.1, section 6</i>	<i>Included in updated PD.</i>	<i>Validation team response:</i> <i>The updated project description (PD version 02) includes the information regarding the local stakeholder consultation undertaken by the PP.</i>

			Validation team conclusion: Closed
CL 3 The formulae used to calculate λ in the spreadsheet "EF Colombia 2009-2011" Sheets "lambda 09," "lambda 10" and "lambda 11" is not in accordance with the tool provisions	Tool to calculate the emission factor for an electricity system, version 03.0.0, clause 56, (d)	Corrected. See updated EF spreadsheets.	Validation team response: In the updated and revised file provided by the PP with the $EF_{grid,OM-adj,y}$ calculations ("EF Colombia 2010-2012.xls"), λ_{2010}, λ_{2011} and λ_{2012} were calculated in accordance with the tool provisions Validation team conclusion: Closed
CL 4 The formulae used to classified the lowcost/must run power plants in the spreadsheet "EF Colombia 2009-2011" Sheet "Gen 2009 – 2011" file 44 is not in accordance with the power plant nature	Tool to calculate the emission factor for an electricity system, version 03.0.0, clause 56, (b)	Corrected. See updated EF spreadsheets.	Validation team response: In the updated and revised file provided by the PP with the $EF_{grid,OM-adj,y}$ calculations ("EF Colombia 2010-2012.xls"), The formulae used to classified the lowcost/must run power plants is in accordance with the power plant nature

			Validation team conclusion: Closed
CL 5 The depreciation used in the spreadsheet "Equity IRR Hydroelectric Project El Edén (4Apr13).xls", sheet "Equity IRR" file 29 does not address the civil works + infrastructure and equipment in separate way	Guidelines on the assessment of investment analysis, version 05.0, clause 5	This has been corrected. In the adjusted financial analysis, the depreciation is made separately. The individual investment costs with different depreciation rates (20 years straight-line for civil works and 10 years straight-line for equipment) were not available at decision making. Therefore the equipment costs as per the proposal (from the company Andritz, see "COTIZACION ANDRITZ COLOMBIA.pdf") were used for equipment costs. Since the faster depreciation of equipment leads to a slightly lower tax rate during the project beginning, the IRR tends to increase slightly and this assumption is conservative. However, the overall effect is not that significant. Further answer: See "Equity IRR Hydroelectric Project El Edén (11Jun13)" sent with this protocol.	Validation team response: The updated version of the spreadsheet with the financial analysis used for additionality assessment (with the depreciation consideration included) was not provided to the lead auditor In a further answer to this finding the PP provided the updated version of the spreadsheet with the financial analysis used for additionality assessment Validation team conclusion: Closed